



GATE SMASHERS

GATE2024

Engineering Mathematics

Topic: Probability and Statistics

Subtopic: Combination And Restricted Combination

LECTURE - 6

EXPLANATION

Combination - Different selection that can be made out of a given number of things taking some or all of them at a time are called combinations.



Number of all combinations of n different things taken x at a time is $C(n, x) = {}^nC_x$

$$\rightarrow {}^nC_x = \frac{n!}{x!(n-x)!} \quad \rightarrow {}^nC_0 = 1 \quad \text{Imp} \quad {}^nC_x = {}^nC_{n-x}$$

Imp $\rightarrow$ if ${}^nC_a = {}^nC_b$ then $a=b$ or $a+b=n$

Note - ${}^nC_x + {}^nC_{x-1} = {}^{n+1}C_x$

$${}^nC_x / {}^nC_{x-1} = \frac{n-x+1}{x}$$

RESTRICTED COMBINATION -

Combinations of n things taken x at a time, when x particular things always occur $= {}^{n-x}C_{x-x}$

Combinations of n things taken x at a time, when x particular things never occur $= {}^{n-x}C_x$

NOTES PREPARED BY: ANKIT ARORA



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EXPLANATION



• eg

i) ${}^{n+1}C_{n-1} = 36$, find n

ii) If ${}^nC_{10} = {}^nC_{12}$ find n ?

iii) if ${}^nC_r + {}^nC_{r-1} + {}^{n+1}C_{r-1} = {}^{10}C_2$ find n ? & r ?